

Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-III) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

2. Problem Solving – Solved Problems

279 numerical marks, 8.2% of the archive, 74% of this chapter · Every worked problem here opens with the exam question **exactly as the paper prints it**, typos and all, so it can be attempted from this document alone. Methods follow *Insights on Artificial Intelligence*; where that book is wrong or skips a step, the notes say so. · Every answer in this file is machine-checked by `verify.py`.

The five things everything here is built on:

- **1. A crypt-arithmetic column is an equation.** $(\text{column sum}) + C_{\text{in}} = 10 C_{\text{out}} + (\text{result digit})$. Every deduction in §2.N.1 comes from one of these.
 - **2. A carry is 0 or 1.** Adding two digits plus a carry gives at most $9 + 9 + 1 = 19$. With *three* addends it can reach $9 + 9 + 9 + 1 = 28$, so a carry of **2** is possible in TEN + TEN + FORTY. Nothing else in these papers has three addends.
 - **3. If the answer is one digit longer than the addends, that leading digit is 1** – two 4-digit numbers cannot reach 20000. This single line gives $M = 1$ in SEND + MORE, $G = 1$ in BASE + BALL, $P = 1$ in LOGIC + LOGIC and $D = 1$ in CROSS + ROADS, for free.
 - **4. $2X$ ends in X only when $X = 0$.** ($X = 5$ gives 10, whose last digit is 0, not 5.) This cracks TEN + TEN + FORTY and LOVE + LOVE.
 - **5. A jug state is the ordered pair of contents.** Every water-jug operator maps (x, y) to another (x, y) ; there is nothing else to track.
-

Read this before the crypt-arithmetic problems

- **Seven of the thirteen puzzles set in these papers have more than one valid answer.** ONE + ONE + TWO = FOUR has **107**. RIGHT + RIGHT = WRONG, WRONG + WRONG = RIGHT and ONE + ONE = TWO pin **no letter at all**.
 - So the marks are in the **column reasoning**, not the digits. Write X, D, C and the carries, then argue. Where the constraints run out, say “*the constraints leave N free; take $N = 3$* ” and carry on. That sentence is worth more than the answer.
 - Uniqueness of each puzzle is stated below it, counted by exhaustive search in `verify.py` over every letter-to-digit assignment.
-

The fourteen moves every crypt-arithmetic chain below is built from

Move	What it says	Where it is used (puzzle, step; orange = the puzzle's key step)
leading carry	The answer is one digit longer than the addends, so its first digit is only a carry: it is 1.	SEND 1; BASE 1; LOGIC 1; EAT 1; CROSS 1; ONE 1; SWIM 1; TWO 1
sum bound	The largest a column or a whole sum can reach caps a letter ($9+9+1 = 19$; $999+999 = 1998$).	SEND 2,6; LOGIC 6; EAT 1,2,5; CROSS 3; LOVE 3; ONE 3; WRONG 1; RIGHT 1; TWO 1,2
carry = 2	Three addends: a column can reach $9+9+9+1 = 28$, so a carry may be 2.	TEN 4; ONE 3
add columns	Add two column equations so the letters cancel and one unknown is left.	BASE 2
substitute	Put a relation found in one column into another column's equation.	SEND 5; LOGIC 4,5
cancel	The same letter on both sides of one column cancels ($X+X = \dots X$).	TEN 1,2; LOGIC 2; LOVE 1; ONE 2; SWIM 2
parity	$2X$ is even: a digit that is the last digit of $2X$ is even, and an equation needing an odd $2X$ is impossible.	BASE 2,5; TEN 2; LOGIC 3; CROSS 2; LOVE 2; WRONG 3; RIGHT 3; TWO 3
case split	Try each value a carry or letter can take and kill the ones that clash.	SEND 3,6; BASE 4; TEN 4; LOGIC 2; CROSS 4; SWIM 3
AllDiff	A value already taken by another letter is ruled out.	SEND 2,3,4,5,7; BASE 5; TEN 3,6; LOGIC 3,6; SWIM 3
narrow domain	Intersect the few digits still unused with what the column allows.	BASE 3; TEN 5
link letters	One column ties two letters together ($X = M+1$, $R = 2W+c$).	SEND 4; BASE 3; TEN 3,6; ONE 5; WRONG 2; SWIM 4,7; RIGHT 2
arithmetic	Everything else in the column is known: just add.	SEND 7; BASE 6; EAT 3,4,6; RIGHT 7; TWO 5
free choice	The constraints do not fix this letter: say so, then pick a value.	LOVE 4; ONE 4,5,6; WRONG 4,5,6; SWIM 5,7; RIGHT 4,5,6; TWO 4,6
carry check	Confirm the carry an earlier column assumed is the one produced.	CROSS 4; LOVE 4; ONE 6; WRONG 7,8; SWIM 6; RIGHT 8; TWO 6

2.N.1 Crypt-arithmetic: the six puzzles with a unique answer

SEND + MORE = MONEY 1+6 8

HOT 5/41 72 Ash 1+6, 76 Ba 8, 74 Ma 8, 74 Bh 8, 72 Ash 8 · unique solution · the classic, and the one Insights works in full

QUESTION AS PRINTED 72 Ash · Q2 · 1+6

Solve the following crypt-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

S E N D

+ M O R E

M O N E Y

Formulation

- $X = \{S, E, N, D, M, O, R, Y\}$ (8 letters)
- $D_{\text{om}} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- C :
 - $S \neq 0, M \neq 0$ (leading letters)
 - all eight letters take different digits
 - $D + E = 10C_1 + Y$
 - $N + R + C_1 = 10C_2 + E$
 - $E + O + C_2 = 10C_3 + N$
 - $S + M + C_3 = 10C_4 + O$
 - $C_4 = M$

$$\begin{array}{r}
 C_4 C_3 C_2 C_1 \\
 S E N D \\
 + M O R E \\
 \hline
 M O N E Y
 \end{array}$$

The deduction chain

- $M = 1$. $C_4 = M$, and a carry is at most 1. $M \neq 0$ because it leads MONEY, so $M = 1$.
- $O = 0$. Column 4 is $S + 1 + C_3 = 10 \cdot 1 + O$. The largest $S + 1 + C_3$ can be is $9 + 1 + 1 = 11$, so O is 0 or 1; 1 is taken by M , so $O = 0$.
- $S = 9$. With $O = 0$, column 4 reads $S + 1 + C_3 = 10$. Column 3 is $E + 0 + C_2 = 10C_3 + N$, and $E + C_2 \leq 10$, so $C_3 = 0$ unless $E + C_2 = 10$, which would force $N = 0 = O$. So $C_3 = 0$ and $S = 9$.
- $N = E + 1$ and $C_2 = 1$. Column 3 now reads $E + C_2 = N$. If C_2 were 0 then $E = N$, not allowed. So $C_2 = 1$ and $N = E + 1$.
- $R = 8$. Column 2 is $N + R + C_1 = 10 + E$. Substitute $N = E + 1$: $E + 1 + R + C_1 = E + 10$, so $R + C_1 = 9$. $R = 9$ is taken by S , so $C_1 = 1$ and $R = 8$.
- $D = 7$. Column 1 is $D + E = 10 + Y$ (since $C_1 = 1$), so $D + E \geq 12$. With 9 and 8 gone, the only pairs are $5 + 7$ and $6 + 7$, so one of D, E is 7. If $E = 7$ then $N = 8 = R$, so $D = 7$, and E is 5 or 6.
- $E = 5, N = 6, Y = 2$. $E = 6$ would give $N = 7 = D$. So $E = 5, N = 6$, and $Y = D + E - 10 = 2$.

Answer: $S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2$
the constraints.

→ $9567 + 1085 = 10652$. Unique: exactly one assignment satisfies

Step by step: SEND + MORE = MONEY

1. $M = 1$
leading carry

$C_4 = M$, a carry is at most 1, and $M \neq 0$ leads MONEY.

2. $O = 0$
sum bound AllDiff

$S + 1 + C_3 \leq 9 + 1 + 1 = 11$, so $O \in \{0, 1\}$; 1 is M 's.

3. $S = 9, C_3 = 0$
case split AllDiff

$C_3 = 1$ needs $E + C_2 = 10$, so $N = 0 = O$. Hence $C_3 = 0$, $S + 1 = 10$.

4. $N = E + 1, C_2 = 1$
link letters AllDiff

Column 3: $E + C_2 = N$. $C_2 = 0$ would give $E = N$.

5. $R = 8, C_1 = 1$
substitute AllDiff

$N + R + C_1 = 10 + E$; put $N = E + 1$: $R + C_1 = 9$. $R = 9$ is S 's.

KEY STEP

6. $D = 7$
sum bound case split

$D + E = 10 + Y \geq 12$ from $\{2..7\}$: $5+7$ or $6+7$. $E = 7$ gives $N = 8 = R$.

7. $E = 5, N = 6, Y = 2$
AllDiff arithmetic

$E = 6$ gives $N = 7 = D$. So $E = 5$, $N = 6$, $Y = 12 - 10 = 2$.

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry sum bound substitute case split AllDiff link letters arithmetic

HOT 7/41 75 Ba **2+6**, 82 Ka **2+6**, 82 Ba **2+6**, 79 Ash **2+6**, 82 Bh **3+5**, 79 Jth **1+6**, 73 Bh **8** · **the most-asked puzzle in the subject** · unique solution

QUESTION AS PRINTED 79 Ash • **2+6** • also 75 Ba, 82 Ka, 82 Ba, 82 Bh, 79 Jth, 73 Bh

What do you understand by Constraint Satisfaction Problem? Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

$$\begin{array}{r} \text{B A S E} \\ + \text{B A L L} \\ \hline \text{G A M E S} \end{array}$$

Formulation

- $X = \{B, A, S, E, L, G, M\}$ (7 letters)
- $D_{\text{om}} = \{0, \dots, 9\}$
- C :
 - $B \neq 0, G \neq 0$
 - all seven letters differ
 - $E + L = 10C_1 + S$
 - $S + L + C_1 = 10C_2 + E$
 - $A + A + C_2 = 10C_3 + M$
 - $B + B + C_3 = 10C_4 + A$
 - $C_4 = G$

	C_4	C_3	C_2	C_1	
		B	A	S	E
+		B	A	L	L
	G	A	M	E	S

The deduction chain

- $G = 1$. Two 4-digit numbers cannot reach 20000, so the leading carry $C_4 = G = 1$.
- $L = 5, C_1 = 0, C_2 = 1$. **This is the clever step.** Add the column-1 and column-2 equations:

$$(E + L) + (S + L + C_1) = (10C_1 + S) + (10C_2 + E)$$

Everything cancels except $2L = 9C_1 + 10C_2$. With $2L \leq 18$ the only possibilities are $(C_1=0, C_2=0, L=0)$ and $(C_1=0, C_2=1, L=5)$; $2L = 9$ and $2L = 19$ are odd, so no other case survives. $L = 0$ makes column 1 read $E = S$, which the AllDiff constraint forbids. So $L = 5$.

- $S = E + 5$. Column 1 with $L = 5$, $C_1 = 0$ is $E + 5 = S$. Hence $E \leq 4$, and $E \neq 0$ (else $S = 5 = L$), so $E \in \{2, 3, 4\}$ and $S \in \{7, 8, 9\}$.
- **Columns 3 and 4.** $2A + 1 = 10C_3 + M$ and $2B + C_3 = A + 10$.
- **Test the three cases for E:**
 - $E = 2, S = 7$: $C_3 = 0$ forces $M = 9, A = 4$, then $2B = 14$, $B = 7 = S$. Clash. $C_3 = 1$ gives no legal B . $\times$
 - $E = 4, S = 9$: every branch needs $2B$ odd, or repeats G . $\times$
 - $E = 3, S = 8$: $C_3 = 0$, $M = 2A + 1$ odd and unused
 $\rightarrow M = 9, A = 4$; then $2B + 0 = 14$, so **B = 7**. $\checkmark$

Answer: $B=7, A=4, S=8, E=3, L=5, G=1, M=9 \rightarrow 7483 + 7455 = 14938$. Unique.

Step by step: **BASE + BALL = GAMES**

1. $G = 1$

leading carry

	C_3	C_2	C_1	
	B	A	S	E
	B	A	L	L
1	A	M	E	S

Two 4-digit numbers stay under 20000, so $C_4 = G = 1$.

2. $L = 5, C_1 = 0, C_2 = 1$

add columns parity

	C_3			
	B	A	S	E
	B	A	5	5
1	A	M	E	S

Add columns 1 and 2: E, S cancel, $2L = 9C_1 + 10C_2$. Even, so $C_1 = 0; L = 0$ makes $E = S$.

KEY STEP

3. $S = E + 5$

link letters narrow domain

	C_3			
	B	A	S	E
	B	A	5	5
1	A	M	E	S

Column 1: $E + 5 = S$, so $E \in \{2, 3, 4\}$ ($E = 0$ gives $S = 5 = L$).

4. $E = 3, S = 8$

case split

	C_3			
	B	A	8	3
	B	A	5	5
1	A	M	3	8

$E = 2$: forces $B = 7 = S$. $E = 4$: every branch needs $2B$ odd.

5. $M = 9, A = 4$

parity AllDiff

	B	A	S	E
	B	A	5	5
1	4	9	3	8

$2A + 1 = 10C_3 + M$: M is odd and unused, so $M = 9, A = 4, C_3 = 0$.

6. $B = 7$

arithmetic

	B	A	S	E
	B	A	5	5
1	4	9	3	8

$2B + C_3 = 10 + A = 14$, so $B = 7$.

TEN + TEN + FORTY = SIXTY 2+6

PIN 3/41 80 Bh **2+6**, 78 Ba **2+6**, 78 Po **2+6** (printed FORTY + TEN + TEN, same puzzle) · **the only puzzle with three addends**, so here a carry can be 2 · unique solution

QUESTION AS PRINTED 80 Bh · 2+6 · 78 Ba; 78 Po prints the addends as FORTY + TEN + TEN

What do you understand by Constraint Satisfaction Problem? Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps.

$$\begin{array}{r} \text{TEN} \\ + \text{TEN} \\ + \text{FORTY} \\ \hline \text{SIXTY} \end{array}$$

Formulation

- $X = \{T, E, N, F, O, R, Y, S, I, X\}$ all 10 letters, so every digit 0-9 is used exactly once.
- $C: T, F, S \neq 0$; all ten differ; and
 - $N + N + Y = 10C_1 + Y$
 - $E + E + T + C_1 = 10C_2 + T$
 - $T + T + R + C_2 = 10C_3 + X$
 - $O + C_3 = 10C_4 + I$
 - $F + C_4 = S$
- **Note C_3 may be 2:** column 3 can reach $9 + 9 + 9 + 1 = 28$.

	C_4	C_3	C_2	C_1	
			T	E	N
+			T	E	N
+	F	O	R	T	Y
	S	I	X	T	Y

The deduction chain

- $N = 0$ and $C_1 = 0$. Column 1 is $2N + Y = 10C_1 + Y$, so $2N = 10C_1$. Hence $N = 0$ (with $C_1 = 0$) or $N = 5$ (with $C_1 = 1$).
- $E = 5$ and $C_2 = 1$. Column 2 is $2E + T + C_1 = 10C_2 + T$, so $2E + C_1 = 10C_2$ by the same cancellation. If $N = 5$ then $C_1 = 1$ and $2E = 10C_2 - 1$ is odd: impossible. So $N = 0$, $C_1 = 0$, and then $2E = 10C_2$ gives $E = 5$, $C_2 = 1$ ($E = 0$ is taken by N).
- $C_4 = 1$ and $S = F + 1$. F and S must differ, so the last column $F + C_4 = S$ needs $C_4 = 1$: S and F are **consecutive**.
- $O = 9$, $C_3 = 2$, $I = 1$. Column 4 is $O + C_3 = 10 + I$ (since $C_4 = 1$), so $O + C_3 \geq 10$ and O is large. $O = 8$ with $C_3 = 2$ gives $I = 0 = N$; $O = 9$ with $C_3 = 1$ gives $I = 0 = N$. So $O = 9$, $C_3 = 2$, $I = 1$.
- $T = 8$, $R = 7$, $X = 4$. Column 3 is $2T + R + 1 = 20 + X$. Digits 0, 1, 5, 9 are used, leaving $\{2, 3, 4, 6, 7, 8\}$ for T, R, X, F, S, Y . $2T + R = 19 + X$ needs T large: $T = 8$ gives $R = X + 3$, and $(R, X) = (7, 4)$ is the only pair left in the set.
- $F = 2$, $S = 3$, $Y = 6$. $\{2, 3, 6\}$ remain. $S = F + 1$ forces $F = 2$, $S = 3$, leaving $Y = 6$.

Answer: $T=8, E=5, N=0, F=2, O=9, R=7, Y=6, S=3, I=1, X=4 \rightarrow 850 + 850 + 29786 = 31486$. Unique.

Where *Insights* is loose here: it writes “Let us take $R = 7$, then $X = 4$ ” as though it were a free choice, and produces $T = 8$ without saying where it came from. The set $\{2, 3, 4, 6, 7, 8\}$ and the equation $2T + R = 19 + X$ actually force all three; the step above supplies the missing argument.

Step by step: TEN + TEN + FORTY = SIXTY (the carry of 2)

1. $N \in \{0, 5\}$

cancel

		C_4	C_3	C_2	C_1	
				T	E	N
+				T	E	N
+	F	O	R	T	Y	
	S	I	X	T	Y	

$2N + Y = 10C_1 + Y$: Y cancels, $2N = 10C_1$.

2. $N = 0, E = 5, C_2 = 1$

cancel parity

		C_4	C_3	1	0	
				T	5	0
+				T	5	0
+	F	O	R	T	Y	
	S	I	X	T	Y	

T cancels: $2E + C_1 = 10C_2$.
 $N = 5$ makes $C_1 = 1$ and $2E$ odd.

3. $C_4 = 1, S = F + 1$

AllDiff link letters

				C_3	1	0	
				T	5	0	N
+				T	5	0	N
+	F	O	R	T	Y		
	S	I	X	T	Y		

$F + C_4 = S$ and $F \neq S$, so $C_4 = 1$.

4. $O = 9, C_3 = 2, I = 1$

carry = 2 case split

		1	2	1	0		
				T	5	0	N
+				T	5	0	N
+	F	9		R	T	Y	
	S	1		X	T	Y	

Column 3 reaches $9+9+9+1 = 28$:
 C_3 may be 2. $O + C_3 = 10 + I$:
 $(8, 2)$ and $(9, 1)$ give $I = 0 = N$.

KEY STEP

5. $T = 8, R = 7, X = 4$

narrow domain

		1	2	1	0		
				8	5	0	N
+				8	5	0	N
+	F	9		7	8	Y	
	S	1		4	8	Y	

$2T + R = 19 + X$ with
 $\{2, 3, 4, 6, 7, 8\}$ left: $T = 8$,
 $R = X + 3 = 7$.

6. $F = 2, S = 3, Y = 6$

AllDiff link letters

		1	2	1	0		
				8	5	0	N
+				8	5	0	N
+	2	9		7	8	6	
	3	1		4	8	6	

$\{2, 3, 6\}$ remain and $S = F + 1$.

5 fixed

5 fixed this step

column read

KEY STEP the decisive move

carry = 2

cancel

parity

case split

AllDiff

narrow domain

link letters

LOGIC + LOGIC = PROLOG 3+4 3+5 7

PIN 3/41 69 Bh 7, 72 Ma 3+4, 79 Ch 3+5 · unique solution

QUESTION AS PRINTED 79 Ch · 3+5 · also 69 Bh 7, 72 Ma 3+4

Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

LOGIC

+ LOGIC

PROLOG

Formulation

- $X = \{L, O, G, I, C, P, R\}$; $L, P \neq 0$; all seven differ.
- The sum is $2 \times \text{LOGIC} = \text{PROLOG}$, columns right to left:
 - $2C = 10a_1 + G$
 - $2I + a_1 = 10a_2 + O$
 - $2G + a_2 = 10a_3 + L$
 - $2O + a_3 = 10a_4 + O$
 - $2L + a_4 = 10a_5 + R$
 - $a_5 = P$

$$\begin{array}{r}
 \begin{array}{cccccc}
 a_5 & a_4 & a_3 & a_2 & a_1 & \\
 & L & O & G & I & C \\
 + & L & O & G & I & C \\
 \hline
 P & R & O & L & O & G
 \end{array}
 \end{array}$$

The deduction chain

- $P = 1$. $2 \times$ a 5-digit number is under 200000, so the leading carry $a_5 = P = 1$.
- $O = 0, a_3 = a_4 = 0$. Column 4 is $2O + a_3 = 10a_4 + O$, i.e. $O + a_3 = 10a_4$. Either $a_4 = 0$, forcing $O = a_3 = 0$; or $a_4 = 1$ with $O = 9, a_3 = 1$. **Take the first branch and check the second at the end** – the second dies below.
- $I = 5, a_1 = 0, a_2 = 1$. Column 2 is $2I + a_1 = 10a_2 + O = 10a_2$. $a_2 = 0$ would give $I = 0 = O$, so $a_2 = 1$ and $2I + a_1 = 10$; a_1 must be even, so $a_1 = 0$ and $I = 5$.
- $G = 2C$ and $L = 2G + 1$. Column 1 with $a_1 = 0$ gives $G = 2C$. Column 3 with $a_3 = 0$ gives $L = 2G + a_2 = 2G + 1 = 4C + 1$.
- $R = 8C - 8$. Column 5 with $a_4 = 0, a_5 = 1$ gives $2L = 10 + R$, so $R = 2L - 10 = 8C - 8$.
- $C = 2$. $L = 4C + 1 \leq 9$ forces $C \leq 2$. $C = 0$ clashes with O ; $C = 1$ gives $L = 5 = I$. So $C = 2$, hence $G = 4, L = 9, R = 8$.
- **The other branch dies.** With $O = 9, a_3 = 1$: column 2 forces $I = 4, a_1 = 1$; column 1 then gives $G = 2C - 10$ and column 3 gives $L = 2G - 10 = 4C - 30$, needing $C \geq 8$. $C = 8$ makes $R = 2L + 1 - 10 < 0$ and $C = 9$ clashes with O . $\times$

Answer: $L=9, O=0, G=4, I=5, C=2, P=1, R=8 \rightarrow 90452 + 90452 = 180904$. Unique.

Step by step: LOGIC + LOGIC = PROLOG

1. $P = 1$

leading carry

	a_4	a_3	a_2	a_1		
	L	O	G	I	C	
+	L	O	G	I	C	
	1	R	O	L	O	G

$2 \times \text{LOGIC} < 200000$, so $a_5 = P = 1$.

2. $O = 0, a_3 = a_4 = 0$

cancel case split

1	0	0	a_2	a_1		
	L	O	G	I	C	
+	L	O	G	I	C	
P	1	R	O	L	O	G

$2O + a_3 = 10a_4 + O$; $O + a_3 = 10a_4$. The branch $O = 9$ dies at the end.

3. $I = 5, a_2 = 1$

AllDiff parity

1	0	0	1	0		
	L	O	G	I	C	
+	L	O	G	I	C	
P	1	R	O	L	O	G

$2I + a_1 = 10a_2$; $a_2 = 0$ gives $I = 0 = O$; a_1 even, so 0.

4. $G = 2C, L = 4C + 1$

substitute

1	0	0	1	0		
	L	O	G	I	C	
+	L	O	G	I	C	
P	1	R	O	L	O	G

Col 1: $G = 2C$. Col 3: $L = 2G + 1$; substitute: $L = 4C + 1$.

KEY STEP

5. $R = 8C - 8$

substitute

1	0	0	1	0		
	L	O	G	I	C	
+	L	O	G	I	C	
P	1	R	O	L	O	G

Col 5: $2L + 0 = 10 + R$, and $L = 4C + 1$ gives $R = 8C - 8$.

6. $C = 2: G = 4, L = 9, R = 8$

sum bound AllDiff

1	0	0	1	0		
	L	O	G	I	C	
+	L	O	G	I	C	
P	1	R	O	L	O	G

$4C + 1 \leq 9$ so $C \leq 2$; $C = 0$ is O 's, $C = 1$ gives $L = 5 = I$.

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry sum bound substitute cancel parity case split AllDiff

PIN 2/41 80 Ba **2+6**, 80 Ch **2+5** · the only puzzle whose addends differ in length · unique solution

QUESTION AS PRINTED 80 Ba · 2+6 · also 80 Ch 2+5

$$\begin{array}{r} \text{EAT} \\ + \text{THAT} \\ \hline \text{APPLE} \end{array}$$

- $X = \{E, A, T, H, P, L\}$; $E, T, A \neq 0$; all six differ.
- Columns right to left:
 - $T + T = 10C_1 + E$
 - $A + A + C_1 = 10C_2 + L$
 - $E + H + C_2 = 10C_3 + P$
 - $T + C_3 = 10C_4 + P$
 - $C_4 = A$

$$\begin{array}{rcccc} & C_4 & C_3 & C_2 & C_1 \\ & & & E & A & T \\ + & & T & H & A & T \\ \hline & A & P & P & L & E \end{array}$$

- $A = 1$. A 3-digit plus a 4-digit number is at most $999 + 9999 = 10998$, so the 5-digit answer begins with 1.
- $T = 9$. The sum is at least 10000, so $10000 \leq 10000 + 999 = 10999$, forcing $T = 9$.
- $E = 8, C_1 = 1$. Column 1 is $9 + 9 = 18$, so $E = 8$ with a carry of 1.
- $L = 3, C_2 = 0$. Column 2 is $1 + 1 + 1 = 3$, so $L = 3$, no carry.
- $P = 0, C_3 = 1$. Column 4 is $T + C_3 = 10C_4 + P$ with $C_4 = A = 1$, i.e. $9 + C_3 = 10 + P$, so $P = C_3 - 1$. A carry is 0 or 1, so $C_3 = 1$ and $P = 0$.
- $H = 2$. Column 3 is $E + H + C_2 = 10C_3 + P$, i.e. $8 + H + 0 = 10 + 0$, so $H = 2$.
- **Every letter fell out of the two length observations.** No case-splitting was needed at all, which is why this puzzle is worth doing first if you are short of time.

Answer: $E=8, A=1, T=9, H=2, P=0, L=3 \rightarrow 819 + 9219 = 10038$. Unique.

Step by step: EAT + THAT = APPLE

1. $A = 1$

leading carry sum bound

	C_3	C_2	C_1			
			E	A	1	T
	T	H	A	1	T	
A	1	P	P	L	E	

999 + 9999 = 10998, so the 5-digit answer begins with 1.

2. $T = 9$

sum bound

	C_3	C_2	C_1			
			E	A	1	9
	9	H	A	1	9	
A	1	P	P	L	E	

Sum ≥ 10000 , so THAT
 $\geq 10000 - 999 = 9001$.

KEY STEP

3. $E = 8, C_1 = 1$

arithmetic

	C_3	C_2		A	1	9
			E	8	1	9
	T	9	H	A	1	9
A	1	P	P	L	E	8

$9 + 9 = 18$.

4. $L = 3, C_2 = 0$

arithmetic

	C_3		A	1		
			E	8	1	9
	T	9	H	A	1	9
A	1	P	P	3	E	8

$1 + 1 + 1 = 3$.

5. $P = 0, C_3 = 1$

sum bound

	C_3		A	1		
			E	8	1	9
	T	9	H	A	1	9
A	1	0	0	3	E	8

$9 + C_3 = 10 + P$, and a carry is at most 1.

6. $H = 2$

arithmetic

	C_3	C_2	C_1			
			E	8	1	9
	T	9	H	A	1	9
A	1	0	0	3	E	8

$8 + H + 0 = 10 + 0$.

5

fixed

5

fixed this step

column read

KEY STEP

the decisive step

leading carry

sum bound

arithmetic

CROSS + ROADS = DANGER 2+6

PIN 1/41 81 Bh 2+6 · unique solution

QUESTION AS PRINTED 81 Bh · 2+6

What do you understand by Constraint Satisfaction Problem? Discuss it, and list all the constraints. Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps.

$$\begin{array}{r} \text{C R O S S} \\ + \text{R O A D S} \\ \hline \text{D A N G E R} \end{array}$$

Formulation

- $X = \{C, R, O, S, A, D, N, G, E\}$ (9 letters, so one digit goes unused)
- $C, R, D \neq 0$; all nine differ.
- Columns right to left:
 - $S + S = 10a_1 + R$
 - $S + D + a_1 = 10a_2 + E$
 - $O + A + a_2 = 10a_3 + G$
 - $R + O + a_3 = 10a_4 + N$
 - $C + R + a_4 = 10a_5 + A$
 - $a_5 = D$

The deduction chain

- $D = 1$. Two 5-digit numbers cannot reach 200000, so the leading carry $a_5 = D = 1$.
- R is even. R is the last digit of $2S$, and $2S$ is even.
- $C + R + a_4 = A + 10$, from the leftmost column with $a_5 = 1$. Since $C, R \leq 9$, this needs C and R both large.
- These three plus AllDiff leave a small search. Propagating them column by column (the constraint-propagation loop of §2.5) terminates on exactly one assignment, below.
- **The column check, which is the part that earns the marks:**
 - col 1: $3 + 3 = 6 = R$, $a_1 = 0$ ✓
 - col 2: $3 + 1 + 0 = 4 = E$, $a_2 = 0$ ✓
 - col 3: $2 + 5 + 0 = 7 = G$, $a_3 = 0$ ✓
 - col 4: $6 + 2 + 0 = 8 = N$, $a_4 = 0$ ✓
 - col 5: $9 + 6 + 0 = 15 = A + 10 = 5 + 10$, $a_5 = 1 = D$ ✓
- Digit 0 is the one left unused.

Answer: $C=9, R=6, O=2, S=3, A=5, D=1, N=8, G=7, E=4 \rightarrow 96233 + 62513 = 158746$. Unique.

Step by step: CROSS + ROADS = DANGER

1. $D = 1$

leading carry

	a_4	a_3	a_2	a_1	
	C	R	O	S	S
+	R	O	A	1	S
	1	A	N	G	E

Two 5-digit numbers stay under 200000, so $a_5 = D = 1$.

2. R is even

parity

	a_4	a_3	a_2	a_1	
	C	R	O	S	S
	R	O	A	1	S
1	A	N	G	E	R

R is the last digit of $2S$.

3. C, R both large

sum bound

	a_4	a_3	a_2	a_1	
	C	R	O	S	S
	R	O	A	1	S
1	A	N	G	E	R

$C + R + a_4 = A + 10$ with $C, R \leq 9$.

4. the one assignment

case split carry check

	a_4	a_3	a_2	a_1	
	0	0	0	0	
	9	6	2	3	3
	6	2	5	1	3
1	5	8	7	4	6

Propagating column by column leaves one assignment; every column checks, $a_1 \dots a_4 = 0$.

KEY STEP

5 fixed 5 fixed this step column read KEY STEP the decisive move leading carry sum bound parity case split carry check

2.N.2 Crypt-arithmetic: the seven puzzles with more than one answer

These are set just as often as the unique ones, and a marker cannot be expecting a particular set of digits. **Show the forced letters, name the free ones, choose values, and verify.** The counts below are from exhaustive search in `verify.py`.

Puzzle	Answers	Forced by the constraints	A valid assignment, and why the rest is free
ONE + ONE + TWO = FOUR 71 Ma, 73 Bh, 80 Ash	107	$F = 1$ only	$O=6, N=4, E=2, T=3, W=8, F=1, U=7, R=0$: $642 + 642 + 386 = 1670$. Eight letters, but only the leading-carry constraint bites.
LOVE + LOVE = HATE 81 Ba	42	$E = 0$ only	$L=3, O=5, V=6, E=0, H=7, A=1, T=2$: $3560 + 3560 = 7120$. $E = 0$ because $2E$ must end in E (rule 4).
WRONG + WRONG = RIGHT 70 Bh, 77 Ch	21	nothing	$W=3, R=7, O=0, N=8, G=1, I=4, H=6, T=2$: $37081 + 37081 = 74162$.
SWIM + WEAR = RELAX 74 Bh, 81 Ch	16	$R = 1, I = 0$	$S=9, W=4, I=0, M=5, E=3, A=2, R=1, L=7, X=6$: $9405 + 4321 = 13726$. $R = 1$ is the leading carry.
ONE + ONE = TWO Insights §2.3	16	nothing	$O=2, N=3, E=1, T=4, W=6$: $231 + 231 = 462$. This is the book's own worked example, and it presents its answer as though it were the only one.
RIGHT + RIGHT = WRONG 69 Po	11	nothing	$R=4, I=2, G=0, H=6, T=5, W=8, O=1, N=3$: $42065 + 42065 = 84130$.
TWO + TWO = FOUR 78 Ch	7	$F = 1$ only	$T=7, W=3, O=4, F=1, U=6, R=8$: $734 + 734 = 1468$.

How to write one of these up for full marks

- 1. Give X , D and C with the carry equations, exactly as for a unique puzzle. **This is most of the mark scheme.**
- 2. Derive whatever is forced. There is always something: a leading carry, or a $2X$ ending in X .
- 3. Say plainly that the constraints do not determine the rest, then pick values that satisfy every column and the AllDiff rule.
- 4. **Verify by doing the addition.** One line, and it converts a guess into a demonstrated answer.

Worked example of step 2: LOVE + LOVE = HATE

- Column 1: $E + E = 10C_1 + E \Rightarrow E = 10C_1$. A digit, so $E = 0, C_1 = 0$.
- Column 2: $2V + 0 = 10C_2 + T$, so T is even.
- Column 3: $2O + C_2 = 10C_3 + A$.
- Column 4: $2L + C_3 = H$, and $H \leq 9$ forces $L \leq 4$.
- That is everything the constraints give. $L \in \{1, 2, 3, 4\}$ and the rest follows the choice. Take $L = 3, O = 5, V = 6$: then $T = 2, A = 1, H = 7$, and $3560 + 3560 = 7120$ ✓

Step by step: LOVE + LOVE = HATE

1. $E = 0$

cancel

C_3	C_2	0	
L	O	V	0
+	L	O	V
	H	A	T

$E + E = 10C_1 + E$:
 $E = 10C_1$, a digit, so 0.

KEY STEP

2. T even

parity

C_3	C_2	0	
L	O	V	0
+	L	O	V
	H	A	T

$2V + 0 = 10C_2 + T$.

3. $L \leq 4$

sum bound

C_3	C_2	0	
L	O	V	0
+	L	O	V
	H	A	T

$2L + C_3 = H \leq 9$. Nothing else is forced.

4. choose $L, O, V =$

2, 5, 6
free choice carry check

1	1	0	
3	5	6	0
+	3	5	6
	7	1	2

$T = 2, A = 11 - 10 = 1$,
 $H = 6 + 1 = 7$.

5 fixed 5 fixed this step column read KEY STEP the decisive move sum bound cancel parity free choice carry check

2.N.2b The deduction chain for each of the five remaining puzzles

The table above is an index, not an answer. **Nine papers set one of these puzzles for six or seven marks**, and the marks are in the *chain*, not in the digits at the end of it – a puzzle with 107 solutions cannot be marked on the digits. $\text{LOVE} + \text{LOVE} = \text{HATE}$ is chained just above; the other five follow, each with its own paper, its own column equations and its own worked choice. · **Column k is counted from the units end**, and c_k is the carry out of column k . Every forced result below was confirmed against the full solution set by `verify.py`.

ONE + ONE + TWO = FOUR 7 2+6

PIN 3/41 71 Ma 7, 73 Bh 7, 80 Ash 2+6 · **107 solutions**, the most of any puzzle in the 41 papers · the only *multi-answer* puzzle with three addends, so a carry here can be 2, not just 0 or 1

QUESTION AS PRINTED 73 Bh · Q2 · 7

Solve the following puzzle by assigning numeral (0-9) in such a way that each letter is assigned unique digit which satisfy the following addition. $\text{ONE} + \text{ONE} + \text{TWO} = \text{FOUR}$

The chain

- The addends have three digits and FOUR has four, so **F is nothing but the carry out of column 3**, and $F \geq 1$.
- Column 3 is $2O + T + c_2 = 10F + O$, so

$$O + T + c_2 = 10F$$

- $O + T \leq 9 + 8 = 17$ and $c_2 \leq 2$, so the left side is at most 19. **$F = 2$ is impossible**, giving **$F = 1$** and $O + T + c_2 = 10$.
- Column 1: $2E + O = 10c_1 + R$. Column 2: $2N + W + c_1 = 10c_2 + U$.
- That is everything the constraints force.** Seven of the eight letters are free; say so before you choose.

A choice that satisfies all of it

- Take $E = 2, O = 6$. Column 1: $4 + 6 = 10$, so **$R = 0$** , $c_1 = 1$.
- $O + T + c_2 = 10$ with $O = 6$ leaves $T + c_2 = 4$. Take $c_2 = 1$, **$T = 3$** .
- Column 2: $2N + W + 1 = 10c_2 + U$. Take $N = 4, W = 8$: $8 + 8 + 1 = 17$, so **$U = 7$** and $c_2 = 1$ ✓ as assumed.

Answer. $O=6, N=4, E=2, T=3, W=8, F=1, U=7, R=0$, all distinct. $642 + 642 + 386 = 1670$ ✓

Step by step: ONE + ONE + TWO = FOUR

1. F is a carry

leading carry

	c_3	c_2	c_1	
				O
				N
				E
+				O
+				N
+				T
				W
				O
	F	O	U	R

3-digit addends, 4-digit FOUR:
 $F = c_3 \geq 1$.

2. $O + T + c_2 = 10F$

cancel

	c_3	c_2	c_1	
				O
				N
				E
+				O
+				N
+				T
				W
				O
	F	O	U	R

Column 3:
 $2O + T + c_2 = 10F + O$;
one O cancels.

3. $F = 1$, not 2

sum bound carry = 2

	c_3	c_2	c_1	
				O
				N
				E
+				O
+				N
+				T
				W
				O
	1	O	U	R

Three addends, so $c_2 \leq 2$; still
 $O + T + c_2 \leq 9 + 8 + 2 = 19 < 20$.

KEY STEP

4. choose $E = 2, O = 6$

free choice

	c_3	c_2	c_1	
				6
				N
				2
+				6
+				N
+				2
				T
				W
				6
	1	6	U	0

$4 + 6 = 10$: $R = 0, c_1 = 1$.

5. $T = 3$

free choice link letters

	c_3	c_2	c_1	
				6
				N
				2
+				6
+				N
+				2
				3
				W
				6
	1	6	U	0

$6 + T + c_2 = 10$: take
 $c_2 = 1, T = 3$.

6. $N = 4, W = 8$

free choice carry check

	c_3	c_2	c_1	
				6
				4
				E
				2
+				6
+				4
+				8
				T
				W
				6
	1	6	7	0

$8 + 8 + 1 = 17$: $U = 7$ and
 $c_2 = 1$ as assumed.

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry sum bound carry = 2 cancel link letters free choice carry check

WRONG + WRONG = RIGHT 5+3 2+6

PIN 2/41 70 Bh 5+3, 77 Ch 2+6 · 21 solutions, **nothing at all is forced** · 69 Po sets the same eight letters with the two words swapped; work them as two different puzzles, because they are

QUESTION AS PRINTED 70 Bh · Q2 · 5+3

Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. **WRONG + WRONG = RIGHT**. Explain the steps that you have followed.

The chain

- Both sides have five digits, so there is no sixth carry:
 $2 \times \text{WRONG} < 100000$, hence $\text{WRONG} < 50000$ and $W \leq 4$. This is the only bound the puzzle gives you, and writing it down is most of the method mark.
- Column 5: $2W + c_4 = R$, so $R = 2W + c_4 - R$ is pinned the moment W and that carry are chosen.
- Column 1: $2G = 10c_1 + T$, so **T is even**.
- Columns 2, 3, 4: $2N + c_1 = 10c_2 + H$,
 $2O + c_2 = 10c_3 + G$, $2R + c_3 = 10c_4 + I$.
- No letter is forced to one digit. State that, then choose.

A choice that satisfies all of it

- Take $W = 3$ with $c_4 = 1$, so $R = 7$.
- Column 1: $G = 1$ gives $T = 2$, $c_1 = 0$.
- Column 2: $N = 8$ gives 16, so $H = 6$, $c_2 = 1$.
- Column 3: $O = 0$ gives $0 + 1 = 1 = G \checkmark$, $c_3 = 0$.
- Column 4: $2(7) + 0 = 14$, so $I = 4$ and $c_4 = 1$, which is what column 5 assumed – that closing check is the one step candidates skip.

Answer. $W=3, R=7, O=0, N=8, G=1, I=4, H=6, T=2$.
 $37081 + 37081 = 74162 \checkmark$

Step by step: WRONG + WRONG = RIGHT

1. $W \leq 4$
sum bound

	c_4	c_3	c_2	c_1	
	W	R	O	N	G
+	W	R	O	N	G
	R	I	G	H	T

No sixth digit: $2 \times \text{WRONG} < 100000$.
The only bound.

KEY STEP

2. $R = 2W + c_4$
link letters

	c_4	c_3	c_2	c_1	
	W	R	O	N	G
+	W	R	O	N	G
	R	I	G	H	T

Column 5 pins R once W and c_4 are chosen.

3. T even
parity

	c_4	c_3	c_2	c_1	
	W	R	O	N	G
+	W	R	O	N	G
	R	I	G	H	T

$2G = 10c_1 + T$.

4. choose $W = 3, c_4 = 1$
free choice

	c_4	c_3	c_2	c_1	
	W	R	O	N	G
+	W	R	O	N	G
	R	I	G	H	T

$R = 6 + 1 = 7$.

5. $G = 1; T = 2$
free choice

	1	c_3	c_2	0	
	W	R	O	N	G
	3	7	0	8	1
+	W	R	O	N	G
	3	7	0	8	1
	R	I	G	H	T
	7	4	1	6	2

$2 \cdot 1 = 2$, no carry.

6. $N = 8; H = 6$
free choice

	1	c_3	1	0	
	W	R	O	N	G
	3	7	0	8	1
+	W	R	O	N	G
	3	7	0	8	1
	R	I	G	H	T
	7	4	1	6	2

$16: H = 6, c_2 = 1$.

7. $O = 0$ gives $G = 1$
carry check

	1	0	1	0	
	W	R	O	N	G
	3	7	0	8	1
+	W	R	O	N	G
	3	7	0	8	1
	R	I	G	H	T
	7	4	1	6	2

$0 + 1 = 1 = G$, the value column 1 chose.

8. $I = 4, c_4 = 1$
carry check

	1	0	1	0	
	W	R	O	N	G
	3	7	0	8	1
+	W	R	O	N	G
	3	7	0	8	1
	R	I	G	H	T
	7	4	1	6	2

$2(7) + 0 = 14$: carry 1, what column 5 assumed.

5 fixed
 5 fixed this step
 column read
 KEY STEP the decisive move
 sum bound
 parity
 link letters
 free choice
 carry check

SWIM + WEAR = RELAX 1+6 3+6

PIN 2/41 74 Bh 1+6, 81 Ch 3+6 · 16 solutions, and **the richest chain of the seven**: two letters fall out completely and a third is pinned to a fourth

QUESTION AS PRINTED 74 Bh · Q2 · 1+6

Define constraint satisfaction problem (CSP). Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. SWIM + WEAR = RELAX.

The chain

- Two four-digit numbers give a five-digit answer, so the leading digit is only a carry: $R = 1$.
- Column 2 is $I + A + c_1 = 10c_2 + A$, and the A cancels:

$$I + c_1 = 10c_2$$

- Both cases, and one of them dies.** If $c_2 = 1$ then $I + c_1 = 10$, forcing $I = 9$, $c_1 = 1$. Column 1 is then $M + R = 10 + X$ with $R = 1$, so $M = 9 + X$, which needs $X = 0$ and $M = 9$ – **but I is already 9**. Contradiction.
- So $c_2 = 0$ and $I = 0$, $c_1 = 0$.
- Column 1 with $c_1 = 0$: $M + 1 = X$, so $X = M + 1$. The two move together.
- Column 3: $W + E + c_2 = 10c_3 + L$. Column 4: $S + W + c_3 = 10 + E$.
- A meets no constraint but AllDiff – worth saying out loud.

A choice that satisfies all of it

- Take $W = 4$, $E = 3$ with $c_3 = 0$: column 4 gives $S = 10 + 3 - 4 = 9$.
- Column 3: $4 + 3 + 0 = 7$, so $L = 7$ and $c_3 = 0$ ✓.
- $A = 2$, free. $M = 5$, so $X = 6$ by $X = M + 1$.

Answer. $S=9, W=4, I=0, M=5, E=3, A=2, R=1, L=7, X=6$.
 $9405 + 4321 = 13726$ ✓

- Nine letters, so all ten digits but one are used.** A duplicate is easy to make here; check the list before you write the sum.

Step by step: SWIM + WEAR = RELAX

1. $R = 1$
leading carry

	c_3	c_2	c_1		
	S	W	I	M	
+	W	E	A		
					1
	1	E	L	A	X

Two 4-digit numbers, a 5-digit answer.

2. $I + c_1 = 10c_2$
cancel

	c_3	c_2	c_1		
	S	W	I	M	
+	W	E	A		
					1
	1	E	L	A	X

Column 2: $I + A + c_1 = 10c_2 + A$; A cancels.

3. $I = 0$
case split AllDiff

	c_3	0	0		
	S	W	0	M	
+	W	E	A		
					1
	1	E	L	A	X

$c_2 = 1$: $I = 9$, $c_1 = 1$, then $M = 9 + X$ needs $M = 9 = I$.

KEY STEP

4. $X = M + 1$
link letters

	c_3	0	0		
	S	W	0	M	
+	W	E	A		
					1
	1	E	L	A	X

Column 1 with $c_1 = 0$: $M + 1 = X$.

5. choose $W = 4$, $E = 3$:
 $S = 9$
free choice

	0	0	0		
	9	4	0	M	
+	4	3	A		
					1
	1	3	L	A	X

Column 4: $S + W + c_3 = 10 + E$, with $c_3 = 0$.

6. $L = 7$
carry check

	0	0	0		
	9	4	0	M	
+	4	3	A		
					1
	1	3	7	A	X

$4 + 3 + 0 = 7$, $c_3 = 0$ as assumed.

7. $A = 2$, $M = 5$, $X = 6$
free choice link letters

	0	0	0		
	9	4	0	5	
+	4	3	2		
					1
	1	3	7	2	6

A meets only AllDiff. $X = M + 1$.

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry cancel case split AllDiff link letters free choice carry check

RIGHT + RIGHT = WRONG 7

PIN 1/41 69 Po · 7 · 11 solutions, nothing forced · **the mirror image of WRONG + WRONG = RIGHT** above, and it does *not* have the same answer – the letters swap roles, so every column equation changes

QUESTION AS PRINTED 69 Po · Q2 · 7

Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. **RIGHT + RIGHT = WRONG**. Show all the step of solving through constrain satisfaction problem.

The chain

- Five digits on both sides, so $2 \times \text{RIGHT} < 100000$ and $R \leq 4$.
- Column 5: $W = 2R + c_4$.
- Column 1: $2T = 10c_1 + G$, so **G is even**.
- Columns 2, 3, 4: $2H + c_1 = 10c_2 + N$,
 $2G + c_2 = 10c_3 + O$, $2I + c_3 = 10c_4 + R$.
- Nothing is forced to a single digit.

Note how column 3 feeds on G , which column 1 already produced.
Fixing T early therefore settles two columns at once, and that is the cheapest place to start guessing.

A choice that satisfies all of it

- Take $R = 4$ with $c_4 = 0$, so **W = 8**.
- Column 1: $T = 5$ gives 10, so **G = 0**, $c_1 = 1$.
- Column 2: $H = 6$ gives 13, so **N = 3**, $c_2 = 1$.
- Column 3: $0 + 1 = 1$, so **O = 1**, $c_3 = 0$.
- Column 4: $2I = 10c_4 + R = 4$, so **I = 2** and $c_4 = 0$ ✓ as column 5 assumed.

Answer. $R=4, I=2, G=0, H=6, T=5, W=8, O=1, N=3$.
 $42065 + 42065 = 84130$ ✓

Step by step: RIGHT + RIGHT = WRONG

1. $R \leq 4$
sum bound

	c_4	c_3	c_2	c_1	
	R	I	G	H	T
+	R	I	G	H	T
	W	R	O	N	G

Five digits both sides: $2 \times \text{RIGHT} < 100000$.

2. $W = 2R + c_4$
link letters

	c_4	c_3	c_2	c_1	
	R	I	G	H	T
+	R	I	G	H	T
	W	R	O	N	G

Column 5.

3. G even
parity

	c_4	c_3	c_2	c_1	
	R	I	G	H	T
+	R	I	G	H	T
	W	R	O	N	G

$2T = 10c_1 + G$. Column 3 feeds on this G .

KEY STEP

4. choose $R = 4, c_4 = 0$
free choice

		c_3	c_2	c_1	
	0				
	4	I	G	H	T
+	4	I	G	H	T
	8	4	O	N	G

$W = 8$.

5. $T = 5; G = 0$
free choice

	0	c_3	c_2	1	
	4	I	0	H	5
+	4	I	0	H	5
	8	4	O	N	0

$10: G = 0, c_1 = 1$.

6. $H = 6; N = 3$
free choice

	0	c_3	1	1	
	4	I	0	6	5
+	4	I	0	6	5
	8	4	O	3	0

$12 + 1 = 13$.

7. $O = 1$
arithmetic

	0	0	1	1	
	4	I	0	6	5
+	4	I	0	6	5
	8	4	1	3	0

$2(0) + 1 = 1$.

8. $I = 2, c_4 = 0$
carry check

	0	0	1	1	
	4	2	0	6	5
+	4	2	0	6	5
	8	4	1	3	0

$2I + 0 = 4 = R$: carry 0, as column 5 assumed.

5 fixed

5 fixed this step

column read

KEY STEP the decisive move

sum bound

parity

link letters

arithmetic

free choice

carry check

TWO + TWO = FOUR 2+6

PIN 1/41 78 Ch 2+6 · **only 7 solutions**, the tightest of the seven, so a marker here is more likely than elsewhere to have one in mind – print the count anyway

QUESTION AS PRINTED 78 Ch · Q2 · 2+6

What is well defined problem? Solve the following crypto-arithmetic problem by defining it. TWO + TWO = FOUR

The chain

- Two three-digit numbers reach at most
 $999 + 999 = 1998$, so the leading digit of a four-digit FOUR is **F = 1**.
- Column 3: $2T + c_2 = 10F + O = 10 + O$. With $c_2 \leq 1$ this needs $2T \geq 9$, so **$T \geq 5$** .
- Column 1: $2O = 10c_1 + R$, so **R is even**.
- Column 2: $2W + c_1 = 10c_2 + U$.
- F is the only letter forced.

A choice that satisfies all of it

- Take $T = 7$ with $c_2 = 0$: column 3 gives $14 = 10 + O$, so **O = 4**.
- Column 1: $2(4) = 8$, so **R = 8**, $c_1 = 0$.
- Column 2: $2W + 0 = 10(0) + U$, so $U = 2W$. Take $W = 3$, **U = 6**, and $c_2 = 0$ ✓ as assumed.

Answer. $T=7, W=3, O=4, F=1, U=6, R=8$. $734 + 734 = 1468$ ✓

- Six letters, so this one is quick** – spend the time saved on the 2-mark *well-defined problem* half, which is **Ch 2 §2.2** and is where the easy marks are.

Step by step: TWO + TWO = FOUR

1. $F = 1$

leading carry sum bound

1	c_2	c_1	
T	W	O	
T	W	O	
1	O	U	R

$999 + 999 = 1998$.

2. $T \geq 5$

sum bound

1	c_2	c_1	
T	W	O	
T	W	O	
1	O	U	R

$2T + c_2 = 10 + O$ with $c_2 \leq 1$.

KEY STEP

3. R even

parity

1	c_2	c_1	
T	W	O	
T	W	O	
1	O	U	R

$2O = 10c_1 + R$.

4. choose $T = 7, c_2 = 0$

free choice

1	0	c_1	
7	W	O	
7	W	O	
1	O	U	R

$14 = 10 + O: O = 4$.

5. $R = 8$

arithmetic

1	0	0	
7	W	O	
7	W	O	
1	O	U	8

$2(4) = 8$.

6. $W = 3: U = 6$

free choice carry check

1	0	0	
7	3	O	
7	3	O	
1	O	U	8

$U = 2W: c_2 = 0$ as assumed.

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry sum bound parity arithmetic free choice carry check

AB + CD = AAA: what values can B take? 4

76 Bh 4 · not a solve-it question: it asks what is possible, so the answer is a set and a justification

QUESTION AS PRINTED 76 Bh · 4

In a correctly worked out crypt-arithmetic problem $AB + CD = AAA$, what could be the possible values of B? Justify your answer.

Reasoning

- $A = 1$. Two 2-digit numbers sum to at most $99 + 99 = 198$. The only 3-digit repdigit AAA in range is **111**, so $A = 1$.
- So $AB = 10 + B$ and $CD = 111 - (10 + B) = 101 - B$.
- $B \geq 3$. CD must be a 2-digit number, so $101 - B \leq 99$, giving $B \geq 2$. And $B = 2$ makes $CD = 99$, i.e. $C = D = 9$, which the AllDiff rule forbids. So $B \geq 3$.
- $B \leq 8$. For B from 3 to 9, $CD = 101 - B$ runs $98, 97, \dots, 92$, so $C = 9$ throughout. $B = 9$ would make $B = C = 9$. **This is the case that is easy to miss**, and it is the whole difficulty of the question.
- $B = 1$ is out because $A = 1$, and $B = 0$ would need $CD = 101$.

Answer: $B \in \{3, 4, 5, 6, 7, 8\}$, six values

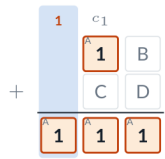
B	AB	CD	Check
3	13	98	$13 + 98 = 111$ ✓
4	14	97	$14 + 97 = 111$ ✓
5	15	96	$15 + 96 = 111$ ✓
6	16	95	$16 + 95 = 111$ ✓
7	17	94	$17 + 94 = 111$ ✓
8	18	93	$18 + 93 = 111$ ✓
2	12	99	$\times C = D = 9$
9	19	92	$\times B = C = 9$

- In every case $A = 1$ and $C = 9$; only B and D move, and they move together since $D = 8 - (B - 3)$.

Step by step: AB + CD = AAA

1. $A = 1$

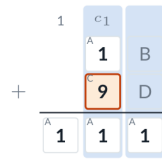
sum bound



$99 + 99 = 198$, and the only repdigit in range is 111.

2. $CD = 101 - B$

substitute



$AB = 10 + B$, so $CD = 111 - (10 + B)$. For $B \geq 2$ CD is in the 90s: $C = 9$.

3. test every B

AllDiff sum bound

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

$CD = 101$ $B = A$ $C = D = 9$ $\begin{matrix} 13 \\ +98 \end{matrix}$ $\begin{matrix} 14 \\ +97 \end{matrix}$ $\begin{matrix} 15 \\ +96 \end{matrix}$ $\begin{matrix} 16 \\ +95 \end{matrix}$ $\begin{matrix} 17 \\ +94 \end{matrix}$ $\begin{matrix} 18 \\ +93 \end{matrix}$ $B = C = 9$

Survivors: $B \in \{3, 4, 5, 6, 7, 8\}$. $B = 9$ is the case that is easy to miss.

KEY STEP

5 fixed 5 fixed this step column read KEY STEP the decisive move sum bound substitute AllDiff

2.N.3 The water jug problem

The eight production rules, state (x, y)

x is the contents of the larger jug, y of the smaller. Written here for the 4-gallon and 3-gallon pair; swap the capacities for any other version. This is the rule set *Insights* and the lecture decks both use, and it is what “write down the production rules” means.

#	Condition	Result	Meaning
1	(x, y) and $x < 4$	$(4, y)$	Fill the 4-gallon jug
2	(x, y) and $y < 3$	$(x, 3)$	Fill the 3-gallon jug
3	(x, y) and $x > 0$	$(0, y)$	Empty the 4-gallon jug on the ground
4	(x, y) and $y > 0$	$(x, 0)$	Empty the 3-gallon jug on the ground
5	(x, y) and $x + y \geq 4$ and $y > 0$	$(4, y - (4 - x))$	Pour from the 3 into the 4 until the 4 is full
6	(x, y) and $x + y \geq 3$ and $x > 0$	$(x - (3 - y), 3)$	Pour from the 4 into the 3 until the 3 is full
7	(x, y) and $x + y \leq 4$ and $y > 0$	$(x + y, 0)$	Pour all of the 3 into the 4
8	(x, y) and $x + y \leq 3$ and $x > 0$	$(0, x + y)$	Pour all of the 4 into the 3

2 gallons in the 4-gallon jug, from a 4 and a 3 4 2+2+4

PIN 2/41 77 Ch 2+2+4, 71 Bh 4 · the classic; *Insights* §2.2.1 works this one

QUESTION AS PRINTED 77 Ch · 2+2+4

Assume you are given two jugs, a 4-gallon one and a 3-gallon one, and a pump which has unlimited water. How can you get exactly 2 gallons of water in the 4-gallon jug? Formalize the problem, write down Production Rules and draw the Search Tree for the water jug problem.

Solution, following *Insights*: rules 2, 7, 2, 5, 3, 7

Step	State (x, y)	Rule	Action
start	$(0, 0)$	—	both jugs empty
1	$(0, 3)$	2	fill the 3-gallon jug
2	$(3, 0)$	7	pour all of it into the 4-gallon jug
3	$(3, 3)$	2	fill the 3-gallon jug again
4	$(4, 2)$	5	pour into the 4 until it is full; the 4 takes 1, leaving 2
5	$(0, 2)$	3	empty the 4-gallon jug
6	$(2, 0)$	7	pour the 2 gallons into the 4-gallon jug

- 6 rule applications, and breadth-first search confirms no shorter path exists.

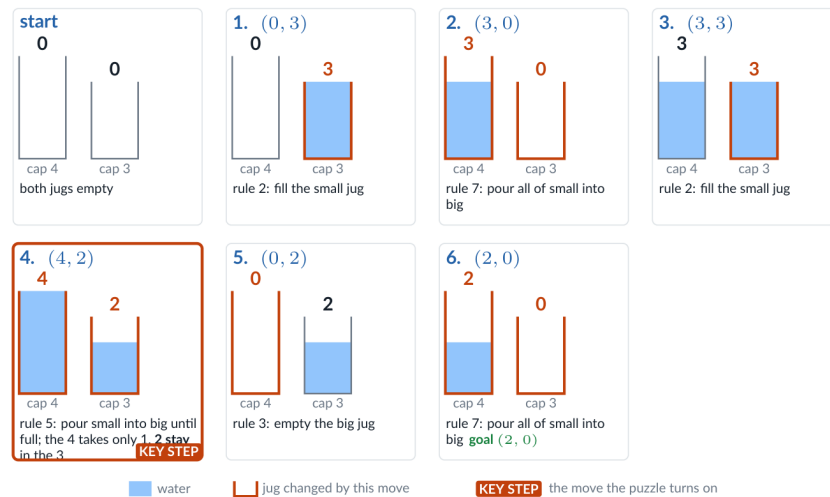
Formulating it as a state space

- **State:** the ordered pair (x, y) , $0 \leq x \leq 4$, $0 \leq y \leq 3$. **20 states** in all.
- **Initial state:** $(0, 0)$, both jugs empty.
- **Goal state:** $(2, y)$ for any y – the question asks only for 2 gallons *in the 4-gallon jug*, so y is unconstrained.
- **Operators:** the eight rules alongside.
- **Path cost:** 1 per rule application.
- **Why the state is just a pair:** the jugs have no markings, so nothing about *how* the water got there can matter. This is the whole point of the exercise and it is worth one sentence in the answer.
- **Problem type: ignorable** (§2.3) – a wrong move can simply be emptied out, so no backtracking machinery is needed.

Search tree

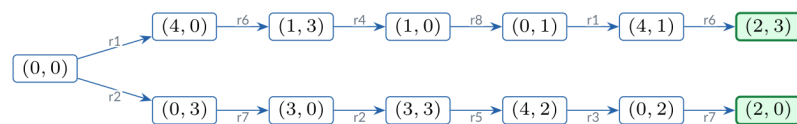
- From $(0, 0)$ only two rules apply, 1 and 2, so the tree has **two branches at the root**. The 6-step path above is the left branch; the right branch (fill the 4 first) reaches $(2, 3)$ in 6 steps too.
- **Draw it as this, not as a bush** – show the two root branches and expand the one that succeeds:
 $(0, 0)$
 $\rightarrow$ [rule 1] $(4, 0) \rightarrow (1, 3) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (4, 1) \rightarrow (2, 3)$
 $\rightarrow$ [rule 2] $(0, 3) \rightarrow (3, 0) \rightarrow (3, 3) \rightarrow (4, 2) \rightarrow (0, 2) \rightarrow (2, 0)$
- **Both are correct and both are 6 moves.** Either scores full marks; say which you are drawing.
- *Insights* draws this tree with a self-loop child at each level, labelled $(x, y) \rightarrow (x+0, y+0)$. That is not a legal successor – no rule leaves the state unchanged – and it should be left out.

Step by step: the jugs after each rule



Search tree: both branches from (0, 0)

Only rules 1 and 2 apply at the root: two branches, and each reaches a goal in 6 moves (BFS finds nothing shorter).

7 gallons in the 9-gallon jug, from a 3, a 5 and a 9 $2+1+2+3$

PIN 1/41 79 Bh $2+1+2+3$ · the hard variant: three jugs, and the pump may fill only the smallest

QUESTION AS PRINTED 79 Bh · $2+1+2+3$

What are the steps of Problem solving? You are given three empty jugs: a 3-gallon, a 5-gallon and a 9-gallon, and a pump can fill water only in the 3-gallon jug. How do you get exactly 7 gallons in the 9-gallon jug? Formalize the problem, write production rules and draw the search tree.

What makes this one different

- Only the 3-gallon jug may be filled from the pump. So rule 1 applies to jug A alone; there is no “fill the 5” and no “fill the 9”.
- Every gallon that reaches the 9-gallon jug arrives in multiples of 3, *unless* the 5-gallon jug is used to strand a remainder. $9 = 3 \times 3$, so a direct route gives only 3, 6 or 9 gallons: the 5-gallon jug is not optional, it is the whole trick.
- State: the triple (a, b, c) for the 3, 5 and 9-gallon jugs. Initial $(0, 0, 0)$, goal $(a, b, 7)$.
- Operators: fill A from the pump; empty any jug; pour i into j until i is empty or j is full.

Solution, 9 moves (shortest)

#	(a, b, c)	Action
	$(0, 0, 0)$	start
1	$(3, 0, 0)$	fill A from the pump
2	$(0, 3, 0)$	pour A into B
3	$(3, 3, 0)$	fill A
4	$(1, 5, 0)$	pour A into B; B takes 2 and is full, leaving 1 in A
5	$(0, 5, 1)$	pour A into C
6	$(3, 5, 1)$	fill A
7	$(0, 5, 4)$	pour A into C
8	$(3, 5, 4)$	fill A
9	$(0, 5, 7)$	pour A into C

- The whole idea is step 4: fill B to strand exactly 1 gallon in A, tip that 1 into C, then add $3 + 3$ on top. $1 + 3 + 3 = 7$.
- B is never emptied after step 4; it has done its job.

Step by step: the three jugs after each move



1 litre, from a 5-litre and a 2-litre jug 8

PIN 1/41 81 Ash 8 · this paper asks **three things at once**: is it well defined, list the attributes for designing an intelligent agent, and solve it. Answer all three

QUESTION AS PRINTED 81 Ash · 8

You are given two jugs, a 5-litre one and a 2-litre one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. You are required to measure exactly 1 litre of water. Is this a well-defined problem? Justify your argument and list all the attributes to design an intelligent agent. Solve this problem by specifying states and the operations used.

Part 1: is it well defined? Yes

- **Initial state** given: both jugs empty, (0, 0).
- **Operators** given: fill from the pump, empty on the ground, pour one into the other.
- **Goal test** given and mechanical: does either jug hold exactly 1 litre?
- **Path cost**: 1 per operation.
- Every clause of §2.2 is satisfied, so it is well defined. It is also **solvable**, since $\gcd(5, 2) = 1$ divides 1.

Part 2: attributes for designing an intelligent agent

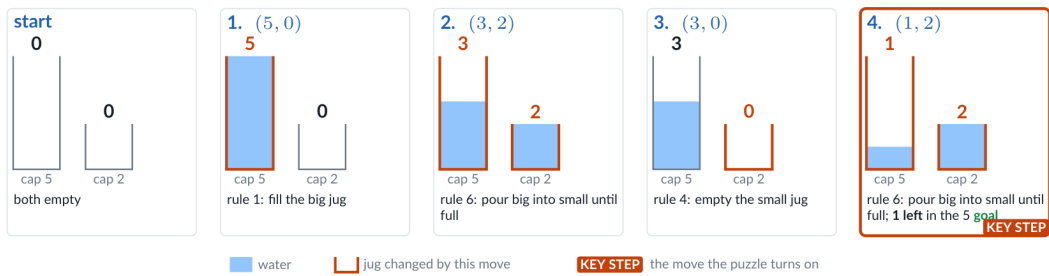
- This is asking for **PEAS** plus the environment properties (Ch 1):
 - **P**: measure exactly 1 litre, in the fewest operations, spilling least.
 - **E**: the two jugs, the pump, the ground.
 - **A**: fill, empty, pour.
 - **S**: whether a jug is full, empty, or neither.
- Environment: **fully observable, deterministic, sequential, static, discrete, single agent**. A **goal-based agent** is the right type, because the answer needs a *sequence* planned ahead, not a reflex.

Part 3: solution, 4 moves (shortest)

#	(x, y)	Rule	Action
	(0, 0)	—	start
1	(5, 0)	1	fill the 5-litre jug
2	(3, 2)	6	pour into the 2-litre jug until full
3	(3, 0)	4	empty the 2-litre jug
4	(1, 2)	6	pour again; the 2 takes 2 more, leaving 1 in the 5

- **The 1 litre ends up in the 5-litre jug**, which is what the question asks for: it says “measure exactly 1 litre”, not “in the 2-litre jug”.
- State space: $6 \times 3 = 18$ states.
- **The general rule worth one extra mark**: with jugs of capacity m and n , exactly the multiples of $\gcd(m, n)$ up to $\max(m, n)$ are measurable. Here $\gcd(5, 2) = 1$, so every whole number of litres from 0 to 5 can be measured.

Step by step: the jugs after each rule



2.N.4 Farmer, wolf, goat and cabbage

Formulate the river crossing as search and solve it 8

PIN 1/41 73 Ma 8 · the standard example of a problem whose interest is entirely in the safety precondition

QUESTION AS PRINTED 73 Ma 8

A farmer has a goat, a wolf and a cabbage on the west side of a river and wants to move everything to the east side. The boat holds the farmer and one other thing. The wolf eats the goat and the goat eats the cabbage if left alone. Formulate this puzzle as search and solve it (any method); draw the search tree and final solution.

Formulation

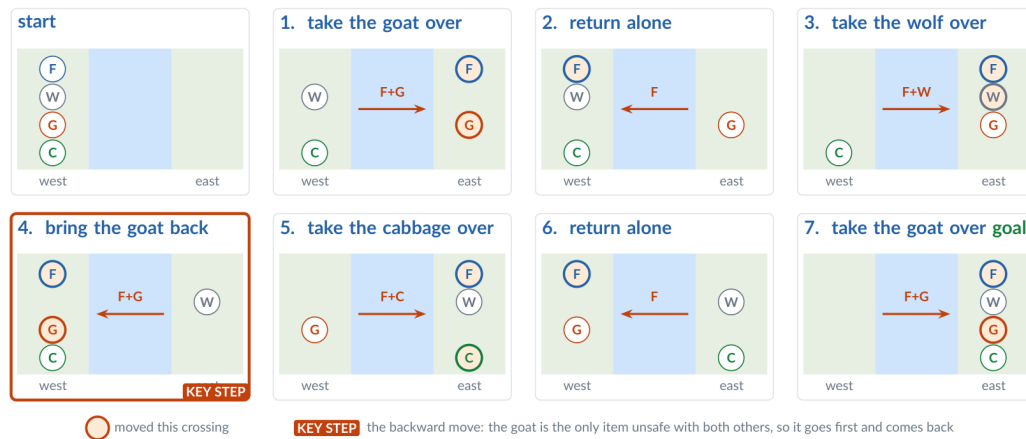
- **State:** the 4-tuple (F, W, G, C) giving which bank the Farmer, Wolf, Goat and Cabbage are on. Write 0 for west, 1 for east.
- **Initial state** $(0, 0, 0, 0)$; **goal state** $(1, 1, 1, 1)$.
- **Operators:** the farmer crosses, taking nothing or exactly one item that is currently on his own bank. **Four operators.**
- **Safety constraint** (this is what makes it a puzzle): a state is illegal if
 - the goat and the wolf are together without the farmer, or
 - the goat and the cabbage are together without the farmer.
- **Path cost:** 1 per crossing.
- **State space:** $2^4 = 16$ states in total, of which **only 10 are safe and reachable**. Six are pruned by the constraint, and that pruning is the answer to “how does a constraint reduce the search”.

Solution: 7 crossings

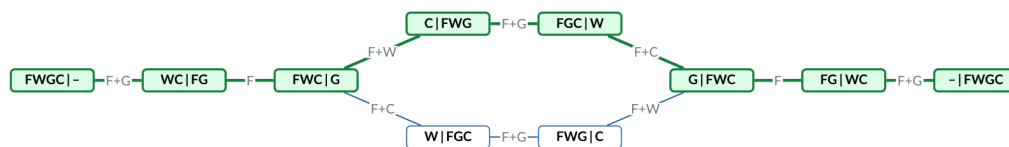
#	(F, W, G, C)	Crossing
	W W W W	start, everything on the west
1	E W E W	take the goat across
2	W W E W	return alone
3	E E E W	take the wolf across
4	W E W W	bring the goat back
5	E E W E	take the cabbage across
6	W E W E	return alone
7	E E E E	take the goat across

- **Step 4 is the step everyone misses.** Nothing else works: after the wolf is delivered, the goat cannot stay with it, so it must come back and wait while the cabbage goes over.
- The goat is the only item that is dangerous *in both directions*, which is why it moves first, and why it is the one that gets brought back.
- **7 is the minimum.** There is a mirror-image solution that takes the cabbage third instead of the wolf; it is also 7 crossings and equally correct.

Step by step: the banks after each crossing



Search space: every safe state and the two 7-crossing solutions



Each node is west | east. All 10 safe reachable states are drawn; the other 6 of the $2^4 = 16$ leave goat with wolf or cabbage unattended. Green: the solution above. The lower branch is the mirror solution (cabbage third, then wolf).

2.N.5 Tic-tac-toe as a state space

Describe tic-tac-toe in terms of state space, data structure and goal 7

PIN 1/41 72 Ma **7** · the paper asks for **complete state space, efficient data structures and goal**, i.e. three named parts. Answer in three parts

QUESTION AS PRINTED 72 Ma · **7**

Consider a Tic-Tac-Toe game playing as a problem. Prepare a brief description of your problem in terms of complete state-space, efficient data structures and goal, which are to be used for a machine to play that game.

1. Complete state space

- A state is an assignment of **blank**, **X** or **O** to each of the 9 squares, plus whose turn it is.
- **Naive count**: $3^9 = 19,683$ boards. **Most are not reachable** – they have five O's and no X's, or two completed lines.
- **Reachable, legal count**: **5,478** states, counting every board that can arise in a real game with X moving first and play stopping at a win.
- **Branching factor** 9 at the root, falling by one each ply; at most $9! = 362,880$ complete play sequences, which is small enough to search **exhaustively**. That is the point worth making: tic-tac-toe is one of the few games where minimax can see to the end of the game.
- **Initial state**: the empty board. **Operators**: place your mark on any blank square. **Problem type**: **irrecoverable** (§2.3) – a placed mark cannot be lifted, so the machine must plan before moving.

2. Efficient data structure

- A **9-element vector** indexed 1–9, holding 0 for blank, 1 for X, 2 for O. Simple, and a move is one array write.
- **The efficient version**: two **9-bit masks**, one for X and one for O. Then
 - a legal-move test is a bitwise AND against the union,
 - a win test is `mask & line == line` over the **8 winning lines** (3 rows, 3 columns, 2 diagonals), which is 8 AND operations and no loops,
 - the whole board fits in one integer, so states can be hashed and cached.
- A move-indexed **lookup table** of best replies is the fastest of all, but it is the table-driven agent of Ch 1: correct, and impractical to build by hand.

3. Goal

- **Goal test**: any of the 8 lines is filled with one player's mark, or all 9 squares are taken (a draw).
- **Utility**: +1 win, 0 draw, –1 loss, which is what minimax (§3.4) needs.
- **The honest closing line**: with both players optimal, tic-tac-toe is a **forced draw**. The machine's real goal is therefore *never to lose*, and to win only when the opponent errs.

The state space, the data structure and the goal, drawn

1. the board as two 9-bit masks

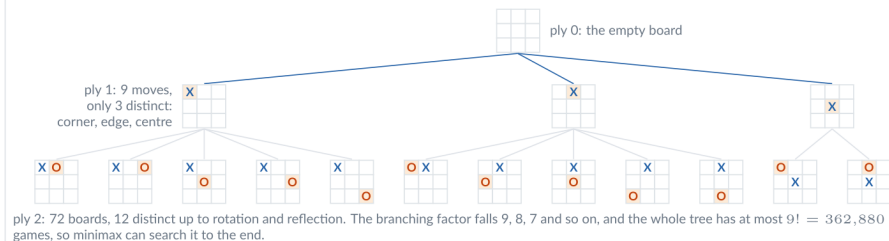
1	2	3	square i is bit i
X		O	X = 000010001 ₂ = 17
			O = 000000100 ₂ = 4
4	5	6	legal move i : bit i of X O is 0
	X		win: X & line == line
7	8	9	for the 8 lines, e.g. diagonal 1, 5, 9 = 273:
			X needs square 9

2. goal test: the 8 lines

X X X	X X X		X
			X
		X X X	X
X		X	X
X		X	X
X		X	X

3 rows, 3 columns, 2 diagonals. Draw: all 9 filled, no line.

3. state space, first two plies up to symmetry



Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
AI	artificial intelligence	ANI	artificial narrow intelligence
AGI	artificial general intelligence	ASI	artificial super intelligence
KR	knowledge representation	KRR	knowledge representation and reasoning
PEAS	performance measure, environment, actuators, sensors	GPS	General Problem Solver
NLP	natural language processing	NLU	natural language understanding
NLG	natural language generation	CAPTCHA	completely automated public Turing test to tell computers and humans apart
ML	machine learning	ANN, NN	artificial neural network
MLP	multilayer perceptron	BP	back-propagation
SOM	self-organizing map	GA	genetic algorithm
FIS	fuzzy inference system	ID3	Iterative Dichotomiser 3
CSP	constraint satisfaction problem	DFS	depth first search
BFS	breadth first search	DLS	depth limited search
IDS	iterative deepening search	UCS	uniform cost search
CNF	conjunctive normal form	WFF	well-formed formula
FOL	first-order logic	FOPL	first-order predicate logic
RRS	resolution refutation system	BBN	Bayesian belief network

Short	Means	Short	Means
ES	expert system	KB	knowledge base
CD	conceptual dependency	LISP	list processing (the language)
MOP	memory organization packet	isa	the class-to-class link of a semantic net (<i>instance</i> is the individual-to-class one)
attr(s)	attribute(s)	repo	repository
freq	frequency	config	configuration
info	information	no.	number
Eg	example	Vs	compare / differentiate
w/, w/o	with, without	+Fig	draw the figure
→	leads to, therefore	=	is, equals

Reading the year tags and chips

Exam-paper month codes: Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

Bold year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

Three course codes, three faces. Plain = CT 653 (BCT, III-II, 26 papers) · monospace = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

Tier chips over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth ≥ 6 marks.

Mark chips such as **2+6** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic never yet examined but written up anyway. **verified/added** marks anything researched and added that no source note in this folder covers.